Ofqual QAN	**600/3055/0**
Quartz ID	**1158**
Qualification	**Level 5 Diploma in Computing**

Assessment	**Network Security and Cryptography 2.0**
Assessment Type	**Examination**
Date	**Wednesday 5th March 2025**
Time	**09.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination

Answer ALL questions

Marks

Question 1

a) Describe the Caesar cipher and encrypt the word **SECURITY** using a Caesar cipher with a key of **FIVE (5)**. **4**

b) Explain the **THREE (3)** core principles of network security in relation to the CIA triad. **6**

Total 10 Marks

Question 2

a) Describe **THREE (3)** methods for securely sharing a secret key used by two parties in symmetric cryptography. **6**

b) Explain the difference between key length and entropy **and** explain how each contributes to ensuring cryptographic security. **4**

Total 10 Marks

Question 3

a) Explain the primary purpose of a Message Authentication Code (MAC) **and** explain the steps involved in generating and verifying a MAC. **6**

b) An employee receives an email from a colleague with an attachment that claims to be an urgent report. However, the email address seems slightly different from the colleague's usual email address. **4**

Identify the type of attack being described **and** suggest **THREE (3)** methods to prevent it from being successful.

Total 10 Marks

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Marks

Question 4

a) Explain the concept of a format string vulnerability **and** discuss **TWO (2)** potential risks associated with this type of vulnerability. **6**

b) Discuss **TWO (2)** advantages of biometric authentication compared to password-based authentication. **4**

Total 10 Marks

Question 5

a) A web-based login system allows users to input their username and password. The application directly uses these inputs to construct an SQL query to verify credentials. An attacker exploits this by entering a specially crafted malicious SQL query in the username field, which is then stored in the system's database. **4**

- Identify the type of vulnerability present in this scenario.
- Explain **TWO (2)** security mitigations that should be implemented to prevent this vulnerability.

b) A company's network has been compromised by ransomware which has spread within its internal network. **6**

Define ransomware **and** discuss **TWO (2)** potential impacts of a ransomware attack on the organisation.

Total 10 Marks

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Marks

Question 6

a) A company aims to secure its email communications to safeguard sensitive information and ensure message authenticity. Recommend and describe **THREE (3)** protocols that can be implemented to achieve these goals. **6**

b) Explain in detail **TWO (2)** good practices for securing web applications against Cross-Site Scripting (XSS) attacks. **4**

Total 10 Marks

Question 7

a) Explain the difference between signature-based intrusion detection **and** anomaly-based intrusion detection systems. **6**

Describe **ONE (1)** advantage **and ONE (1)** disadvantage of each system.

b) What specific elements of a packet does a packet filtering firewall examine to determine whether to allow or block it? **4**

Total 10 Marks

Question 8

a) Company ABC intends to use VPNs (Virtual Private Networks) to ensure secure communication between remote employees and their several branch networks. **6**

Explain how a VPN ensures secure communication **and** describe **TWO (2)** types of VPNs that you will recommend.

b) Explain the difference between a Local Area Network (LAN) **and** a Virtual Local Area Network (VLAN). **4**

Total 10 Marks

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Marks

Question 9

a) Describe the concept of Wireless LAN for Trusted Systems (WLTS) **and** its applications. **4**

b) Company XYZ recently purchased and installed various IoT devices such as smart thermostats and security cameras on their network. **6**

Discuss **THREE (3)** measures the company can take to enhance the security of its network to reduce the risk of vulnerabilities associated with IoT devices.

Total 10 Marks

Question 10

a) Explain the term Risk Management **and** describe the difference between Risk Appetite **and** Risk Tolerance in relation to information security. **6**

b) Discuss the differences between qualitative **and** quantitative risk assessments in information security. **4**

Total 10 Marks

END OF PAPER

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